

1 ☆ QUESTION: 1 GROUP: Aptitude SECTION: Logical Reasoning Mark(s): 1

2 **Direction:** Study the information given below and answer the questions that follow.

3 Six friends Alice, Bob, Charlie, Dana, Emma, and Frank decided to take part in a friendly race. The one who finished first is ranked 1 and so on. After the race, they discussed their placements, but no one remembered
4 exactly where everyone finished. However, they remembered the following clues:

- 5 - Alice finished before Bob but after Charlie.
6 - Emma finished immediately before Charlie.
7 - Frank finished immediately after Dana.
8 - Bob did not finish last.

9 **Question:** Who finished last among the group of friends?

- 10 ☐ Alice
11 ☐ Charlie
12 ☐ Frank
13 ☐ Dana

14

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QUESTION: 2 GROUP: **Aptitude** SECTION: **Logical Reasoning** Mark(s): 1

2 ☆

Direction: Study the information given below and answer the questions that follow.

3

Six friends Alice, Bob, Charlie, Dana, Emma, and Frank decided to take part in a friendly race. The one who finished first is ranked 1 and so on. After the race, they discussed their placements, but no one remembered exactly where everyone finished. However, they remembered the following clues:

4

5

- Alice finished before Bob but after Charlie.

6

- Emma finished immediately before Charlie.

7

- Frank finished immediately after Dana.

8

- Bob did not finish last.

9

Question: Who finished at the first place?

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☐ Charlie

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☐ Alice

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☐ Emma

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☐ Dana

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- 3 ☆
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QUESTION: 3 GROUP: **Aptitude** SECTION: **Logical Reasoning** Mark(s): 1

Direction: Study the information given below and answer the questions that follow.

Six friends Alice, Bob, Charlie, Dana, Emma, and Frank decided to take part in a friendly race. The one who finished first is ranked 1 and so on. After the race, they discussed their placements, but no one remembered exactly where everyone finished. However, they remembered the following clues:

- Alice finished before Bob but after Charlie.
- Emma finished immediately before Charlie.
- Frank finished immediately after Dana.
- Bob did not finish last.

Question: How many people finished between Charlie and Frank?

☐ 2

☐ 3

☐ 4

☐ 1

1

QUESTION: 4 GROUP: Aptitude SECTION: Logical Reasoning Mark(s): 1

2

Direction: Study the information given below and answer the questions that follow.

3

Six friends Alice, Bob, Charlie, Dana, Emma, and Frank decided to take part in a friendly race. The one who finished first is ranked 1 and so on. After the race, they discussed their placements, but no one remembered exactly where everyone finished. However, they remembered the following clues:

4 ☆

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- Alice finished before Bob but after Charlie.

6

- Emma finished immediately before Charlie.

7

- Frank finished immediately after Dana.

8

- Bob did not finish last.

9

Question: If the rankings of Alice and Bob are exchanged, then how many people finished between Emma and Bob?

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☐ 1

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☐ 3

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☐ 4

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☐ 2

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QUESTION: 5 GROUP: **Aptitude** SECTION: **Logical Reasoning** Mark(s): 1

2

Direction: Study the information given below and answer the questions that follow.

3

Six friends Alice, Bob, Charlie, Dana, Emma, and Frank decided to take part in a friendly race. The one who finished first is ranked 1 and so on. After the race, they discussed their placements, but no one remembered exactly where everyone finished. However, they remembered the following clues:

4

5 ☆

- Alice finished before Bob but after Charlie.
- Emma finished immediately before Charlie.
- Frank finished immediately after Dana.
- Bob did not finish last.

6

7

Question: What will be the value of the sum of ranks of Alice and Dana?

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☐ 9

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☐ 7

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☐ 8

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☐ 10

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Previous Question

Next Question

1 QUESTION: 6 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

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Twenty-four friends, along with another person 'X', decide to contribute a sum of money for a road trip. Each of the twenty-four friends contributes ₹30. X agrees to pay ₹16 more than twice the average of the 25 contributors. What is the total amount of money (in ₹) collected by the 25 people?

☐ 800

☐ 780

☐ 720

☐ 840

Previous Question

Next Question

2 QUESTION: 7 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

3 Two persons A and B purchase two types of salts costing Rs 5 per kilogram and Rs. 6 per kilogram respectively. Both of them create a mixture individually and sell the mixture at rupees 7 per kilogram. A makes a profit

4 of 20% and B makes a profit of 25%. If both of them make 30 kilograms of the mixture, then how much kilogram of salt costing Rs. 6 is present in total of 60Kg of mixture?

- 5 ☐ 18 kg
- 6 ☐ 43 kg
- 7 ☆ ☐ 45 kg
- 8 ☐ 25 kg
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QUESTION: 8 GROUP: Aptitude SECTION: Data Analysis Mark(s): 1

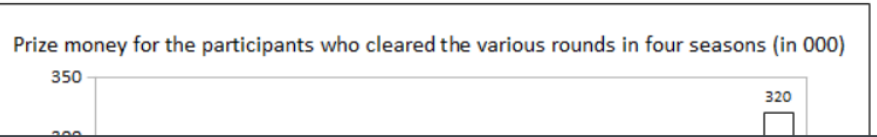
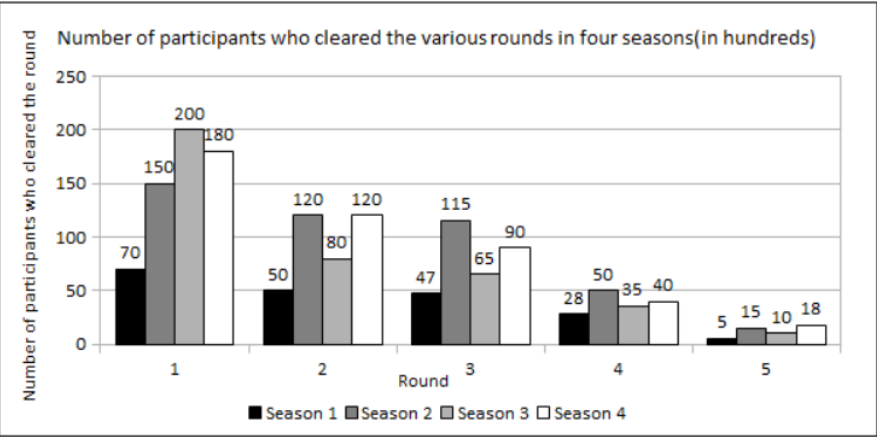
Direction: Read the following information and answer the question that follows.

A XYZ television channel conducted a quiz programme of four seasons, the quiz contest contains five different rounds and they will also give some prize money for successful completion of rounds.

The total number of people who enrolled for the contest of four seasons is 2 lakhs, 3 lakhs, 4 lakhs and 5 lakhs respectively. And for the each season 10%, 15%, 10% and 5% of the people are selected randomly from the enrolled people respectively.

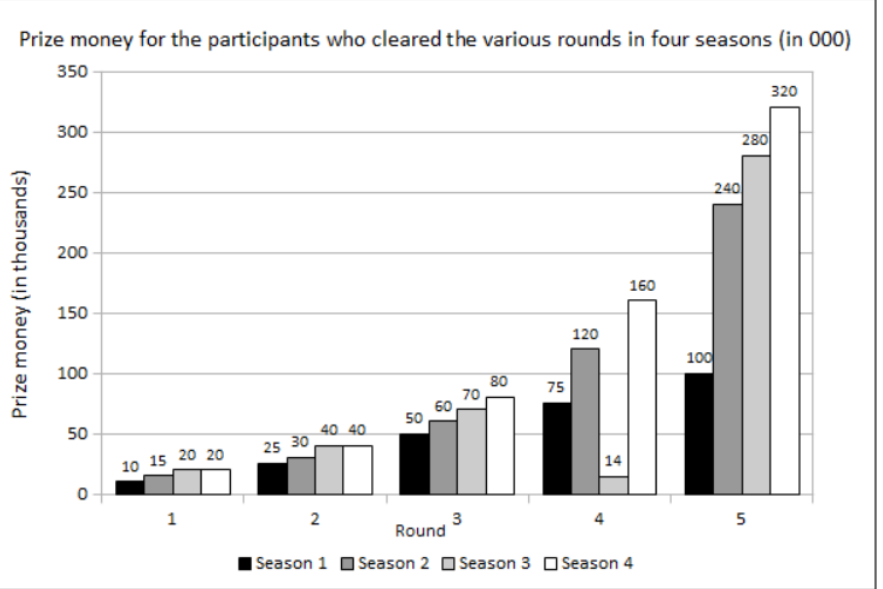
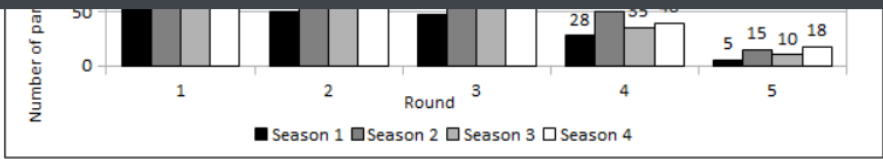
Bar graph 1: Shows the number of participants who cleared various rounds in the contest in each season.

Bar graph 2: Shows the amount of prize money given to the participants who cleared the respective rounds in the contest.



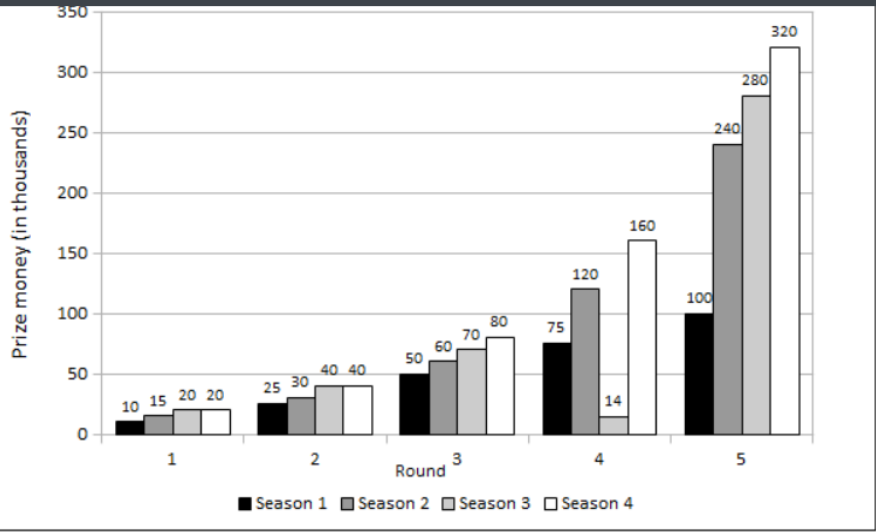
Previous Question

Next Question



Question: If the enrolment fee for each applicant is Rs. 110 for the second season of the contest then find the difference between expenditure and income of the channel for that season on the participants who cleared the first round?

- ☐ Rs. 1500 lakhs
- ☐ Rs. 1850 lakhs



Question: If the enrolment fee for each applicant is Rs. 110 for the second season of the contest then find the difference between expenditure and income of the channel for that season on the participants who cleared the first round?

☐ Rs. 1500 lakhs

☐ Rs. 1850 lakhs

☒ Rs. 1790 lakhs

☐ Rs. 1920 lakhs

QUESTION: 9 GROUP: Aptitude SECTION: Data Analysis Mark(s): 1

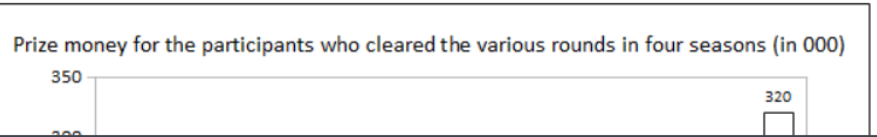
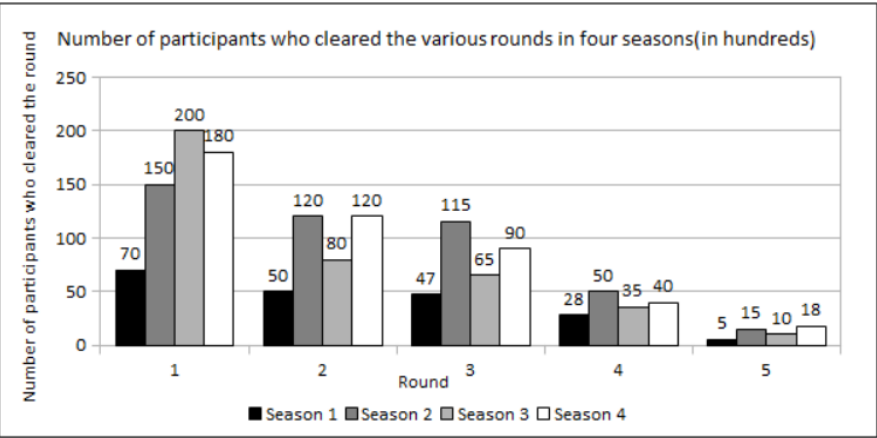
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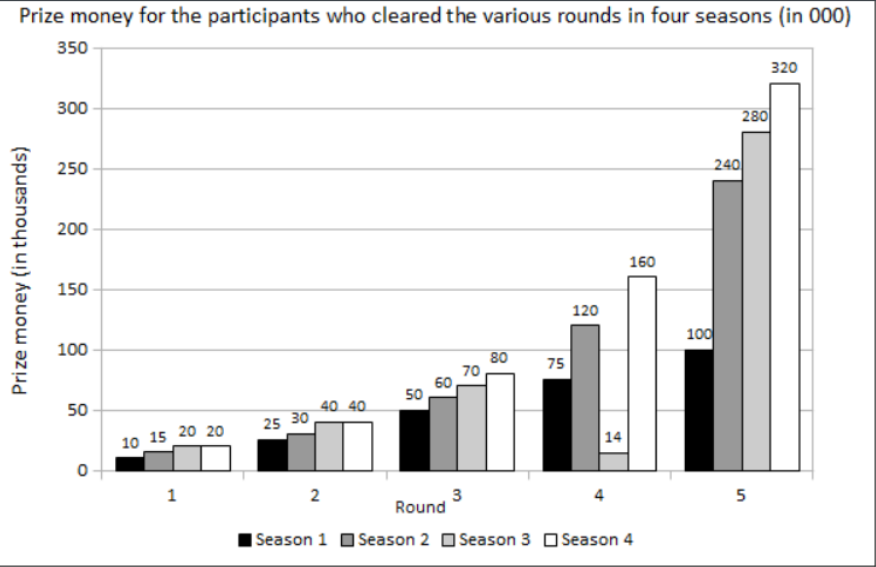
Bar graph 1: Shows the number of participants who cleared various rounds in the contest in each season.

Bar graph 2: Shows the amount of prize money given to the participants who cleared the respective rounds in the contest.



Previous Question

Next Question



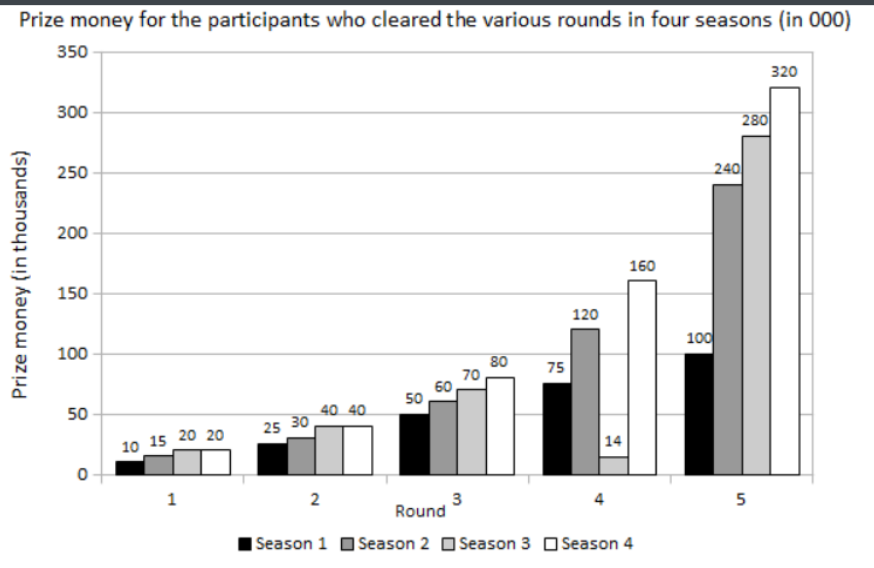
Question: The total income earned by the channel for the fourth season if the enrolment fee is Rs. 120 is what percentage of the expenditure spent on fourth round of that season?

☐ 7.150 %

☐ 8.425 %

☐ 9.375 %

☐ 10.25 %



Question: What is the ratio of number of participants who cleared the fifth round in all the seasons together to the average number of people who were selected for the contest for all the four seasons together?

☐ 48 : 311

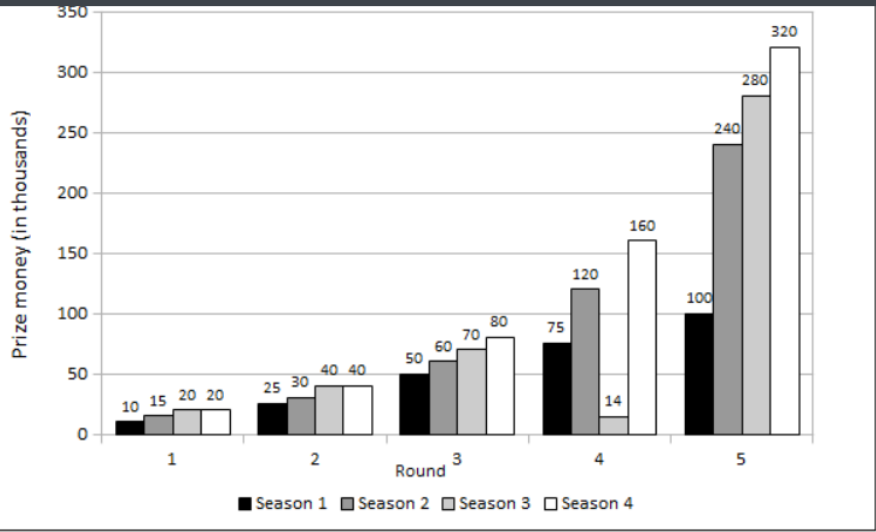
☐ 29 : 325

☐ 19 : 39

☐ 48 : 325

Previous Question

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Question: The number of people who enrolled but not selected for the first season is approximately what percentage of the total number of participants who cleared up to three rounds(first, second & third round) in all the seasons together?

- ☒ 115%
- ☐ 135%
- ☐ 140%
- ☐ 120%

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QUESTION: 12 GROUP: Aptitude SECTION: Puzzles Mark(s): 1

If CLIMB is coded as FOFJE, then which of thee following terms will be coded as VDCBU?

- ☐ None of these
- ☐ SWEAR
- ☐ SAFER
- ☐ SADLY

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13 ☆

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QUESTION: 13 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

If I leave my office at quarter past six and reach the station at half past seven I will be ten minutes earlier than the person who is fifteen minutes late to the train. At what time does the train start?

☐ 7.35 pm

☐ 7.25 pm

☐ 7.40 pm

☐ 7.20 pm

Previous Question

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QUESTION: 14 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

A bag contains 3 red , 4 blue and 6 yellow balls. What is the probability of drawing three balls randomly, so that none of the balls is blue?

☐ 44/143

☐ 67/286

☐ 53/286

☐ 42/143

Previous Question

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- 15 ☆
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QUESTION: 15 GROUP: Aptitude SECTION: Puzzles Mark(s): 1

How many such pairs of letters are there in the word "KEYBOARD" which has as many letters between them (from both the sides) as in the alphabets from A to Z?

- ☐ 1
- ☐ 2
- ☐ 4
- ☐ 3

- 7
- 8
- 9
- 10
- 11
- 12
- 13
- 14
- 15
- 16 ☆
- 17
- 18
- 19
- 20

QUESTION: 16 GROUP: Aptitude SECTION: Puzzles Mark(s): 1

How many degrees will the second - hand move, in the same time in which the minute hand moves 74°?

- ☐ 4420°
- ☐ 4440°
- ☐ 4240°
- ☐ 4540°

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17 ☆

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QUESTION: 17 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

There are 16 vacancies for clerk in a certain office. 20 applications are received. In how many ways can the clerks be selected?

☐ 3876

☐ 969

☐ 3800

☐ 4845

Previous Question

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- 18 ☆
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QUESTION: 18 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

A pipe can fill a tank in 15 hours. Due to a leak in the bottom, it is filled in 20 hours. If the tank is full, how much time will the leak take to empty it?

- ☐ 80 hours
- ☐ 20 hours
- ☐ 40 hours
- ☐ 60 hours

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QUESTION: 19 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

There is square room with adjacent walls AB and BC. A door is placed on point D which is diagonally opposite to point B. the door is opened inside the room. A carpet is placed on the floor of the room in quarter circular form such that the circle centre is at B and radius equals to the side of the square. It touch the door at point P on the circular arc. Find the ratio of the area of the room to area covered by the door.

☐ 2:(2-3√2)

☐ 3:(5-3√2)

☐ 3:(2-3√2)

☐ 2:(3-2√2)

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QUESTION: 20 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

A shopkeeper purchases a box of apples from a wholesale trader. It was found that 20% of the apples were rotten and from the remaining, 5% were not yet ripe. The shopkeeper kept 83.33% of the good ripened fruit and 50% of the rotten fruits on his stall for sale which were 110 in number. Find the total number of unripe apples in the box.

☐ 9

☐ 12

☐ 6

☐ 15

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21 ☆ QUESTION: 21 GROUP: **Programming Basics** SECTION: **C Programming** Mark(s): 1

22 What is the meaning of the statement given below?

23 `Char *ipointer = NULL;`

24 ☐ ipointer is pointing to a garbage value.

25 ☐ ipointer is pointing to nothing.

26 ☐ It means a invalid declaration.

27 ☐ ipointer is pointing to the value "zero".

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29 [Next Question](#)

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21 QUESTION: 22 GROUP: Programming Basics SECTION: Operating System Fundamentals Mark(s): 1

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Which of the statement(s) given below is/are correct?

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I) When a job is assigned to a queue in multilevel queue scheduling, it can change the queue type over course of time.

24

II) In multilevel queue scheduling, it is necessary to implement scheduling between multiple queues too.

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☐ Neither (I) nor (II)

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☐ Both (I) and (II)

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☐ Only (I)

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☐ Only (II)

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Previous Question

Next Question

21 QUESTION: 23 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

22 What will be the output of the program given below?

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```
#include <stdio.h>

int i = 10, j = 15;

void Sample(int **q, int **p)
{
    int l = **q;
    *q = *p;
    **p = l;
    **p *= **q;
    **q += **p;
};

int main()
{
    int *p = &i, *q = &j;
    Sample(&p, &q);
    printf("%d %d", i, j);
    return 0;
}
```

33

☐ 10 0

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☐ 10 150

Previous Question

Next Question

21 QUESTION: 24 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

22 What will be the output of the program given below?

23 `#include <stdio.h>`
24 ☆ `int main()`
25 `{`
26 `int x = 5, *p;`
27 `p = &x;`
28 `printf("%d", ++*p);`
29 `return 0;`
30 `}`

☐ compile time error

☐ 6

☐ Garbage value

☐ 5

Previous Question

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21 QUESTION: 25 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

22 What will be the value of "p" in the code snippet given below?

23 `short int p = 4;`
24 `++p;`
25 `p <<= p >> 2;`

- 26 ☐ 10
- 27 ☐ 16
- 28 ☐ 8
- 29 ☐ 0

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QUESTION: 26 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

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What will be the output of the program given below?

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#include <stdio.h>

25

int main()

26 ☆

{

static int m = 4, n = 2;

m <<= n | 3 & m;

n = n >> 2 | m + 1 << n - 1;

27

printf("%d %d", m, n);

28

return 0;

}

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☐ 4 1

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☐ 16 34

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☐ 16 12

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☐ 0 13

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22 QUESTION: 27 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

23 A one dimensional array A is converted to a two dimensional array N[3][10] using column major order. What will be the address of N[3][3], if base address N[1][1] is 1000 and N[1][2] is at 1012?

24 A: 1 2 3 4 5 6 7 8 9 10 11 12

26 ☐ 1024

27 ☆ ☐ 1032

28 ☐ 1080

29 ☐ 1048

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23 QUESTION: 28 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

24 Which one of the given options is the equivalent prefix expression for the infix expression given below?

25 $(A + B) - (C + D * E) / F * G$

- 26
- 27 ☐ $- + AB * / + CDE * FG$
- 28 ☆ ☐ $- / + AB * + CDE * FG$
- 29 ☐ $/ - + AB * + C * DEFG$
- 30 ☐ $- + AB * / + C * DEFG$
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24 QUESTION: 29 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

25 Consider the three binary numbers given below:

- 26 I) 11011110
27 II) 11011011
28 III) 10101100

29 ☆ What is the value obtained in binary when all the three binary numbers are added?

- 30 ☐ 110110111
- 31 ☐ 1001100100
- 32 ☐ 1001100101
- 33 ☐ 1011100101
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QUESTION: 30 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

26

Consider the time required for the five functional units, that operate in each of the five cycles :

27

20 ns, 16 ns, 20 ns, 20 ns and 14 ns.

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Assume that the pipelining adds 1 ns of overhead. What is the speed up versus single cycle data path for the given units?

30 ☆

☐ 6.0 times

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☐ 4.5 times

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☐ 4.3 times

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☐ 6.4 times

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Previous Question

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26 QUESTION: 31 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

27 What will be the result of converting the number given below in hexadecimal format?

28 $(1234)_{10}$

30 ☐ $(4D2)_{16}$

31 ☆ ☐ $(2D2)_{16}$

32 ☐ $(4D4)_{16}$

33 ☐ $(4D1)_{16}$

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QUESTION: 32 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

Choose the correct option regarding RISC (Reduced Instruction Set Computer) and CISC (Complex Instruction Set Computer) architecture:

- I) RISC architecture has its focus on hardware implementations.
- II) RISC architecture features single clock simple instructions.
- III) CISC architecture has its focus on software implementations.
- IV) CISC architecture features multi-clock simple instructions.
- V) Pipelining is much suitable for RISC architecture as compared to CISC.

☐ Only (III) and (IV)

☐ Only (II) and (V)

☐ Only (I) and (IV)

☐ Only (II), (IV) and (V)

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QUESTION: 33 GROUP: Programming Basics SECTION: Operating System Fundamentals Mark(s): 1

Which of the below given statement(s) is/are correct for best fit?

- I) The smallest hole which is big enough is allocated to program.
- II) The largest hole which is big enough is allocated to program.

☐ Only (II)

☐ Only (I)

☐ Neither (I) nor (II)

☐ Both (I) and (II)

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QUESTION: 34 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

Consider the given below :

- I) Load the result.
- II) Fetch the first operand
- III) Perform the calculation.
- IV) Fetch the instruction.

What is the correct order of the steps for the execution of a complete instruction?

☐ (I), (II), (III) and (IV)

☐ (IV), (II), (III), (I)

☐ (IV), (III), (I), (II)

☐ (II), (III), (IV) and (I)

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QUESTION: 35 GROUP: Programming Basics SECTION: Operating System Fundamentals Mark(s): 1

When two processes communicate with each other, they may use asymmetric way of addressing each other. Which of the given options justifies the syntax of this method of communication?

☐ send (P, message) and receive (Q, message)

☐ send (id, message) and receive (Q, message)

☐ send (id, message) and receive (Q, id, message)

☐ send (P, message) and receive (id, message)

QUESTION: 36 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

Which of the statement given in options is correct about the code snippet given below?

```
#include <stdio.h>

int main(int *q, int qa)
{
    *q = 4, qa = 3;
    int i;
    for(i = 0; i < qa; i++)
        printf("%d", *(q + i) = ++*(q) + i);
    return 0;
}
```

☐ Program will print output as : 5

☐ Program will throw compile time error.

☐ Program will print output as : 4

☐ Program will throw run time error.

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QUESTION: 37 GROUP: Programming Basics SECTION: Error Finding Mark(s): 1

What is the likely cause of the error in the given code below?

```
#include <stdio.h>

int main() {
    int arr[5];
    for (int i = 0; i <= 5; i++) {
        arr[i] = i;
    }
    printf("Array elements: ");
    for (int i = 0; i < 5; i++) {
        printf("%d ", arr[i]);
    }
    return 0;
}
```

☐ Syntax error

☐ Buffer overflow

☐ Memory leak

☐

Previous Question

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QUESTION: 38 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

Consider an empty circular queue of size $N = 6$, what will be the index number of FRONT and REAR after performing the below operations?

- I) Insert 5 items
- II) Delete 4 items

☐ FRONT = 3, REAR = 4

☐ FRONT = 4, REAR = 4

☐ FRONT = 0, REAR = 4

☐ FRONT = 0, REAR = 0

Previous Question

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QUESTION: 39 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

What will be the inorder traversal of the binary search tree given below?

☐ 15 28 30 31 59 67 80

☐ 31 30 59 80 28 67 15

☐ 15 28 67 80 30 31 59

☐ 31 28 67 15 30 59 80

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QUESTION: 40 GROUP: Programming Basics SECTION: Error Finding Mark(s): 1

Identify the line in which the error exists.

- ☐ Line 1
- ☐ Line 6
- ☐ Line 8
- ☐ Line 4

41 ☆ QUESTION: 41 GROUP: Computer Science SECTION: Computer Architecture 1 Mark(s): 1

42 Choose the correct option(s) related to the multiplexed connection method of a bus design:

- 43 I) Fewer bus lines are needed which makes the design economical.
- 44 II) Design of multiplexing is complex.
- 45 III) Performance is reduced.
- 46 IV) This design is also known as dedicated bus design.

- 47 ☐ Only (II) and (III)
- 48 ☐ Only (II)
- 49 ☐ Only (I), (II) and (III)
- 50 ☐ Only (I), (III) and (IV)

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41 QUESTION: 42 GROUP: Computer Science SECTION: C programming Mark(s): 1

42 ☆

What will be the output of the program given below?

43

#include <stdio.h>

44

int swap(int *a, int *b)

45

{
 *a = *a + *b; *b = *a - *b; *a = *a - *b;

46

int main()

47

{

int x = 10, y = 20;

48

swap(&x, &y);

printf("x = %d y = %d\n", x, y);

49

return 0;

50

}

51

☐ x = 20 y = 20

52

☐ Compile time error

53

☐ x = 10 y = 10

54

☐ x = 20 y = 10

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41 QUESTION: 43 GROUP: Computer Science SECTION: OOPS Mark(s): 1

42 Choose the correct statement(s) regarding "Static binding".

- 43 ☆
- 44 I) It happens during compile time.
II) It occurs during class load time.
III) It occurs during runtime.

45 ☐ Only (II)

46 ☐ Only (II) and (III)

47 ☐ Only (I) and (II)

48 ☐ Only (I)

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- 44 ☆
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QUESTION: 44 GROUP: Computer Science SECTION: OOPS Mark(s): 1

When a derived class inherits only from one class and is not further inherited is:

- ☐ hybrid inheritance
- ☐ multilevel inheritance
- ☐ single inheritance
- ☐ multiple inheritance

41 QUESTION: 45 GROUP: Computer Science SECTION: C programming Mark(s): 1

42 What will be the output of the program given below?

```
43 #include <stdio.h>
44 int main()
45 {
46     int a = 4, b = 5;
47     int *m, *n;
48     m = &a; n = &b;
49     if (*m > *n ? *m || *n ? *m : *n : ++*m++)
50         printf("%d", ++*n++);
51     else
52         printf("%d", --*n);
53     return 0;
54 }
```

- 51 ☐ 7
- 52 ☐ 4
- 53 ☐ Garbage value
- 54 ☐ 6

41

QUESTION: 46 GROUP: Computer Science SECTION: Algorithms Mark(s): 1

42

What will be the return value on calling the function func(4, 3)?

43

44

```
int func(int x, int y)
```

45

```
{
```

```
    int count = 0;
```

```
    x = x + 1;
```

46 ☆

```
    if (x >= 0)
```

```
        return x + y;
```

47

```
    else
```

48

```
        return count;
```

```
}
```

49

50

☐ 8

51

☐ 6

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☐ 10

53

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☐ 12

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42 QUESTION: 47 GROUP: Computer Science SECTION: Computer Architecture 1 Mark(s): 1

43 The content of a 4 – bit register is initially 1101. The register is shifted 2 times to the right with the serial input being 1011101. What is the content of the register after each shift?

44 ☐ 1110, 0111

45 ☐ 1001, 1001

46 ☐ 1101, 1011

47 ☆ ☐ 0001, 1000

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QUESTION: 48 GROUP: Computer Science SECTION: C programming Mark(s): 1

44

If the statement "printf("%d", ting(x, y, z));" is placed in place of //blank in the program given below, then what will be the output?

45

46

#include <stdio.h>

static int v, n;

47

int main()

48 ☆

{

int x, *y, **z;

49

x = 'A';

y = &x;

50

z = &y;

51

//blank

return 0;

52

}

53

int ting(int a, int *b, int **c)

{

54

**c += a == 2;

*b += 3;

55

n += 2 << v;

56

return n || v || b;

}

57

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```
43 int x, *y, **z;  
44 x = 'A';  
45 y = &x;  
46 z = &y;  
47 //blank  
48 return 0;  
49 }  
50  
51 int ting(int a, int *b, int **c)  
52 {  
53     **c += a == 2;  
54     *b += 3;  
55     n += 2 << v;  
56     return n || v || b;  
57 }
```

☐ It will print some garbage value.

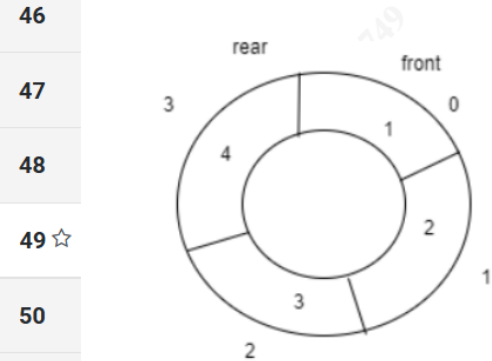
☒ This program will show compile time error.

☐ It will print the output as : 1.

☐ This program will show run time error.

44 QUESTION: 49 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

45 Consider a circular queue of size 4 and initially, the circular queue is full.



52 Consider a queue with the following elements (size 4)

53

5	6	3	2
Front			Rear

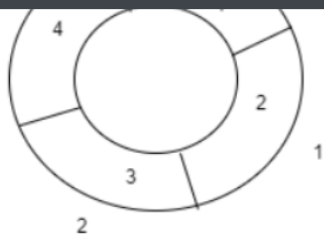
54 Perform the following operation.

- 55
- 1) Dequeue the elements one by one from the circular queue only if the element is less than or equal to the element value in the queue.
 - 2) Insert them into the queue if and only if the value of the circular queue is greater than the value in the queue.

56 Identify the elements present in the queue after performing the specified operation.

57 ☐ 5 6

58 ☐ 2 4



Consider a queue with the following elements (size 4)

5	6	3	2
Front			Rear

Perform the following operation.

1) Dequeue the elements one by one from the circular queue only if the element is less than or equal to the element value in the queue.

2) Insert them into the queue if and only if the value of the circular queue is greater than the value in the queue.

Identify the elements present in the queue after performing the specified operation.

- ☐ 5 6
- ☐ 2 4
- ☐ None of the options given
- ☐ 3 2

45

QUESTION: 50 GROUP: Computer Science SECTION: C programming Mark(s): 1

46

What will be the output of the program given below?

47

48

#include <stdio.h>

int main(void)

49

{

int a = 1.00;

50 ☆

int b = -2.22;

printf("%.2f %d ", --b, --b);

51

printf("%.2f %d", ~a, ~a<<1);

52

return 0;

}

53

☒ 0.00 -4 0.00 -4

54

☐ Compile time error

55

56

☐ -3.22 -4 0.00 -4

57

☐ -3.22 -4 -2.00 -3

58

59

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46 QUESTION: 51 GROUP: Computer Science SECTION: Operating Systems Mark(s): 1

47 Assume the following jobs are to be executed on a single processor system. Jobs arrive at time 0 in the order given below:

48

JOB	CPU Burst Time
M	30
N	50
O	80

49

50

51 ☆ What is the average turnaround time for the jobs, assuming a shortest job first (SJF) scheduling policy?

- 52 ☐ 60 sec
- 53 ☐ 30 sec
- 54 ☐ 90 sec
- 55 ☐ 120 sec
- 56

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52 ☆

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QUESTION: 52 GROUP: Computer Science SECTION: Computer Architecture 1 Mark(s): 1

Consider the below given expression:

```
int v = *z;
```

Which type of addressing is used in the given expression?

☐ Immediate addressing mode

☐ Autoincrement addressing mode

☐ Indexed addressing mode

☐ Indirect addressing mode

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53 ☆
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QUESTION: 53 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

What will be the infix expression for the given postfix expression?

4 3 + 3 / 2 ^ 8 7 + 5 / 3 / 2 * 4 ^ +

- ☐ $((4+3)/3)^2 + (8+7)/(5/3)*2^4$
- ☐ $((4+3)/(3^2)) + (((8+7)/5)/3)*2^4$
- ☐ $((((4+3)/3)^2) + ((8+7)/5)/(3*2)^4$
- ☐ $((4+3)/3)^2 + (((8+7)/5)/3)*2^4$

QUESTION: 54 GROUP: Computer Science SECTION: C programming Mark(s): 1

What will be the output of the program given below?

```
#include <stdio.h>

int main()
{
    int a = 2, b = 3, c = 4;
    b >>= 2 * c++;
    switch(++b * c - a)
    {
        case 1 : b >>= 2 * c++;
                printf("%d", b);
                break;

        case 2 : case 3 : case 4:
                c = ++c * b;
                printf("%d", c);
                break;

        default : c >>= a * b - c;
                printf("%d", c);
                break;
    }
}
```

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```
break;
```

```
case 2 : case 3 : case 4:
```

```
c = ++c * b;
```

```
printf("%d", c);
```

```
break;
```

```
default : c >= a * b - c;
```

```
printf("%d", c);
```

```
break;
```

```
}
```

```
printf(" %d %d %d ", a, b, c);
```

```
return 0;
```

```
}
```

☐ 6 2 1 6

☐ Compile time error

☐ 5 3 1 6

☐ Run time error

QUESTION: 55 GROUP: **Computer Science** SECTION: **C programming** Mark(s): 1

What will be the output of the program given below?

```
#include <stdio.h>

static int a = 10;

int main()
{
    unsigned int a = 20;
    int b = 0x9, c = 2, d = 3;
    printf("%d ", ~(b ^ a) | d++ * a++ & c-- / a * b * c);
    return 0;
}
```

☐ -48

☐ 40

☐ -30

☐ 23

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QUESTION: 56 GROUP: Computer Science SECTION: C programming Mark(s): 1

What will be the output of the program given below?

```
#include <stdio.h>

int main()
{
    char *p = "Kerala", *q = "Ooty", *r = "Mysore", *s = "Shimoga";
    enum loc{Kerala = 1, Ooty = 2, Mysore = 3, Shimoga = 4};
    enum loc dir;
    dir = Shimoga;
    if (dir == Shimoga)
        printf("The best place is %s\n", s);
    return 0;
}
```

☐ The best place is Shimoga

☐ The best place is Kerala

☐ The best place is Ooty

☐ The best place is Mysore

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QUESTION: 57 GROUP: Computer Science SECTION: Operating Systems Mark(s): 1

Which of the given below kernel(s) is/are more suitable for real time programming?

- I) Preemptive kernel
- II) Non-preemptive kernel

☐ Only (II)

☐ Neither (I) nor (II)

☐ Both (I) and (II)

☐ Only (I)

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QUESTION: 58 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

What is the maximum number of node splitting operations that may take place if a B-tree of order 4 is built from scratch by 10 successive insertions?

- ☐ 3
- ☐ 5
- ☐ 6
- ☐ 4

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59 ☆

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QUESTION: 59 GROUP: Computer Science SECTION: OOPS Mark(s): 1

Which of the below given statement(s) is/are true?

- I) An abstract class can be instantiated.
- II) An abstract method must be overridden.

☐ Only (I)

☐ Neither (I) nor (II)

☐ Only (II)

☐ Both (I) and (II)

QUESTION: 60 GROUP: **Computer Science** SECTION: **Operating Systems** Mark(s): 1

Which one of the given options is correct for a request edge in a resource allocation graph(RAG)?

- ☐ The assignment edge is a directed edge from the resource node to a process node.
- ☐ The assignment edge is a directed edge from the process node to a resource node.
- ☐ The assignment edge is a directed edge from the resource node to a resource node.
- ☐ The assignment edge is a directed edge from the process node to a process node.

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60 ☆

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